Hybrid Deterministic/Stochastic Cluster Dynamics: Has the Method Been Extended to Three or More Species?

Review prepared for the EUROFER97 Cluster Dynamics Project

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Contents

1 Introduction

Rate-equation cluster dynamics (RECD) is the standard mean-field framework for tracking the nucleation, growth, and coarsening of point-defect and gas-atom clusters in irradiated materials. Even after the He-content and size-grouping reductions described in earlier sections of this document, the state-space becomes intractable when three or more species co-cluster: vacancies (V), self-interstitial atoms (I or SIA), helium (He), hydrogen (H), and solute atoms (Cu, Mn, Ni, Si, ...) each add one independent cluster-size dimension, and the number of coupled ODEs grows exponentially.

Three broad strategies have been proposed to overcome this *curse of dimensionality*:

1. **Deterministic grouping and Fokker–Planck** methods, which replace the per-size ODE ladder with a coarser representation in one or two size dimensions and remain fully tractable up to two species.
2. **Hybrid deterministic/stochastic** methods, which solve rate equations exactly for small clusters and propagate large clusters via a stochastic (jump-process or Langevin) solver that scales with the number of particles, not the number of species.
3. **Pure stochastic cluster dynamics** (SCD), which recasts the entire master equation as a kinetic Monte Carlo (kMC) problem and avoids the ODE system altogether.

This review focuses on method 2 as introduced by Terrier et al. [1], and asks whether it or its close relatives have been applied to three-species (V + I + He) or higher-dimensional cluster populations. We trace the lineage of methods, report what has and has not been demonstrated, and place the result in the context of the broader multispecies CD literature.

2 The Terrier et al. (2017) Hybrid Method

2.1 Core algorithm

Terrier et al. [1] proposed a generic coupling algorithm that exploits a Lie–Trotter operator splitting of the full RECD system into two sub-dynamics alternating on a macro-timestep δt:

1. **Vacancy/monomer sub-dynamics.** The concentration of the mobile monomer (C_{vac}) evolves with all cluster concentrations $(C_n)_{n\geq 2}$ held fixed. This sub-problem is a scalar nonlinear ODE that can be solved analytically or in the quasi-stationary limit ($\mathrm{d}C_{\mathrm{vac}}/\mathrm{d}t \approx 0$).
2. **Cluster sub-dynamics.** The cluster concentrations $(C_n)_{n\geq 2}$ evolve with C_{vac} held fixed. A key observation is that this sub-problem is *linear* in the C_n, so the distributions of small and large clusters can be propagated *independently and in parallel.*

The cluster sub-dynamics is further split at a frontier size N_{front}: small clusters ($n < N_{\mathrm{front}}$) are integrated by a standard implicit ODE scheme; large clusters ($n \geq N_{\mathrm{front}}$) are evolved by one of two stochastic methods:

- **Jump-process approach.** The forward-Kolmogorov equation for the cluster-size distribution is interpreted as a time-dependent Markov jump process. Independent stochastic particles are propagated, each jumping up or down in size space.
- **Langevin process approach.** For clusters large enough that the Fokker–Planck (FP) approximation is valid, the distribution is propagated via an Euler–Maruyama scheme applied to the associated stochastic differential equation.

The total error in an observable of interest scales as

$$\eta_2^{\text{tot}}(t) \approx \varepsilon_1(t)\,\delta t + \frac{\varepsilon_2(t)}{N_{\text{part}}}, \tag{1}$$

where the first term is the first-order Lie–Trotter splitting error and the second is the statistical sampling error ($\propto N_{\text{part}}^{-1/2}$ by the central limit theorem).

2.2 Demonstrated application

The proof of concept in Ref. [1] is a **single-species** model: vacancy clustering during thermal ageing of a nickel-like metal, parameterised from the GMIC++ benchmark of Ovcharenko et al. [2]. The authors explicitly state that, to their knowledge, no system with three or more defect species had been treated with a deterministic approach up to large cluster sizes at the time of publication, which was the primary motivation for designing the stochastic large-cluster step to be dimensionally scalable.

2.3 Advantages of the coupling

Eliminates the stiffness bottleneck for large clusters. In a purely stochastic simulation (e.g., the Gillespie SSA), the timestep is set by the fastest event in the system. The characteristic jump time $\nu(n)^{-1} = (\beta_n C_{\text{vac}} + \alpha_n)^{-1}$ is $\sim 10^{-2}$ s for $n = 2$ but ~ 10 s at $n = N_{\text{front}} = 200$. By handling small clusters deterministically (where implicit ODE solvers absorb the stiffness) and large clusters stochastically (where the jump frequency is manageable), the hybrid avoids the $\sim 10^3\times$ slowdown that pure SCD suffers from high-frequency small-cluster events.

Circumvents the curse of dimensionality for multiple species. A purely deterministic FP approach works well in 1D (one species) and with some effort in 2D (e.g., He–vacancy pairs), but becomes prohibitive for three or more species: the mesh-based PDE discretisation requires $O(N_{\text{mesh}}^d)$ unknowns, growing exponentially with the number of cluster-size dimensions d. Stochastic particle methods scale as $O(N_{\text{part}})$ regardless of d, because each particle is a point in size-space rather than a mesh node.

Near-linear parallel scaling. Because the cluster concentration sub-problem is linear when C_{vac} is fixed, the trajectories of the stochastic particles are independent of each other. Steps (M1) and (M2) of the algorithm can therefore be executed simultaneously on separate processor cores, giving near-ideal parallel speedup up to the deterministic small-cluster serial bottleneck.

Controlled, quantified error with two independently tunable parameters. The two error contributions (δt for the splitting error; N_{part} for the statistical error) can be balanced independently, giving the user direct control over accuracy versus computational cost.

2.4 Conditions under which benefits are realised

The hybrid offers the greatest advantage when:

- The cluster-size distribution spans many decades ($N \sim 10^3$–10^6), making per-size ODE integration prohibitive.
- Three or more defect/solute species co-cluster, making deterministic mesh-based FP intractable.
- The system is in a growth or coarsening regime so that C_{vac} is quasi-stationary ($\mathrm{d}C_{\mathrm{vac}}/\mathrm{d}t \ll C_{\mathrm{vac}}/T$).
- High-performance parallel hardware is available.
- Small-cluster jump frequencies greatly exceed large-cluster frequencies (the case in most irradiation scenarios).

The benefit is reduced or absent when:

- The system is in the nucleation-dominated regime where C_{vac} changes rapidly, forcing small δt that erode the speedup.
- Only one or two species are involved and $N \lesssim 500$: a well-implemented logarithmic grouping [3, 4] or the high-order FP scheme of Jourdan et al. [5] is simpler, deterministic, and sufficient.
- Spatial correlations from individual displacement cascades matter; in that case, object kinetic Monte Carlo (OKMC) is preferred.
- N_{part} is too small: statistical noise in the large-cluster distribution feeds back into the C_{vac} update and can destabilise the vacancy dynamics.

3 The Deterministic Fokker–Planck–Rate-Equation Coupling (Jourdan 2014)

The direct predecessor of the Terrier 2017 hybrid, and the method that established the CEA/EDF production code CRESCENDO, is the deterministic FP–rate-equation coupling of Jourdan, Bencteux, and Adjanor [6]. This couples standard ODE integration for small clusters to a finite-volume discretisation of the FP equation for large clusters, using the Kurganov–Noelle–Petrova (KNP) scheme with MP5 reconstruction [5] to suppress numerical diffusion in the advection term.

The most advanced application of the deterministic coupling is a **two-species** (V + He) problem: helium implantation in α-iron followed by annealing at high temperature, studying the effect of free surfaces on the coarsening of He bubbles [6]. Extending to three species was explicitly identified as intractable within the deterministic framework, because the FP equation becomes a ($d \geq 3$)-dimensional PDE and mesh-based methods suffer exponential growth in the number of discretisation unknowns. This is the direct motivation for the stochastic large-cluster step introduced in Terrier et al. [1].

4 Three-Species and Multispecies Applications: Only via Pure SCD

Extensive searching of the post-2017 literature reveals that the Terrier hybrid has not been extended to three or more species in any published work as of 2025. The multispecies problem has instead been solved exclusively by **pure stochastic cluster dynamics** (SCD) methods from the Marian/Bulatov group at UCLA/LLNL.

4.1 Three-species (V + He + H): Marian and Bulatov (2011)

Marian and Bulatov [7] introduced SCD as a fully stochastic alternative to standard RECD. Rather than solving a continuous ODE system for defect concentrations in an infinite volume, SCD evolves an integer-valued defect population in a finite material volume Ω, sampling from the underlying kinetic master equation via the Gillespie SSA. Computational complexity is therefore set by the size of the stochastic population, not by the dimensionality of the cluster space.

The key application in Ref. [7] was the first-ever simulation of **triple-beam irradiation** of model ferritic alloys with heavy Fe ions, He, and H simultaneously, carried to 1 dpa. The authors explicitly state this represents the first dpa/He/H simulation in an ion-irradiated ferritic alloy. This three-species (V + He + H) simulation was not feasible by any deterministic approach at the time.

4.2 Enhanced three-species SCD: Hoang et al. (2015)

Hoang, Marian, Bulatov, and Hosemann [8] introduced three algorithmic improvements to SCD specifically to enable high-dose multispecies simulations:

1. **Dynamic reaction-network updating:** the set of active reactions is updated on-the-fly as new cluster species appear, allowing the reaction network to expand without pre-enumeration.
2. **τ-leaping:** a variant of the Gillespie algorithm that advances the reaction network in large time increments τ when the propensities change slowly, trading some accuracy for large speedups in the growth regime.
3. **Volume rescaling:** the stochastic population is periodically rescaled to control the computational cost of the expanding reaction-network, while maintaining accurate statistics.

The enhanced SCD was applied to iron thin films under triple ion-beam irradiation (Fe^{3+}, He^{+}, H^{+}), for which the paper states that standard rate theory or spatially-resolved kMC simulations are prohibitively expensive. This remains the most comprehensive published demonstration of three-species CD to high dose.

4.3 Multispecies SCD for W–H: hydrogen retention in damaged tungsten

The SCD method has also been applied to hydrogen penetration and trapping in pre-damaged tungsten, a multispecies problem involving V clusters, SIA clusters, and H clusters in a plasma-facing context. This again exploits SCD's ability to treat the combinatorial cluster space without pre-specifying the reaction network.

4.4 Radiation-induced precipitation with > 3 species: an open problem

No published application of the Terrier hybrid, the Jourdan FP coupling, or pure SCD to radiation-induced precipitation involving four or more species (e.g., V + I + Mn + Ni + Si + Cu in reactor pressure vessel steels) has been found in the literature. The combinatorial explosion for a 4+ species cluster space remains a fundamental open problem. Object kinetic Monte Carlo (OKMC) and atomistic kinetic Monte Carlo (AKMC) remain the tools of choice for such problems, with the limitation that they cannot reach the dpa-scale doses relevant to long-term RPV embrittlement.

5 A Very Recent Development: Logarithmic-Complexity SSA (2025)

A new development from the same CEA/CERMICS group (Jourdan, Athènes et al.) is a stochastic simulation algorithm with *logarithmic* complexity in the number of reactions [9]. The algorithm uses one binary heap per reaction class, exploiting partial propensities and internal times to reduce the algorithmic complexity from $O(N_{\text{reactions}})$ for the direct Gillespie method to $O(\log N_{\text{immobile}})$ per event, where N_{immobile} is the number of distinct immobile cluster species. The complexity is linear in the number of *mobile* species (typically small: one or two), making it ideally suited to models where only monomers or small clusters are mobile.

The demonstrated application is copper precipitation in an FeCu 1% alloy under irradiation (V + Cu: two species), with speedups of several orders of magnitude over the direct Gillespie method. The authors explicitly state that the method paves the way for efficient multispecies simulations. As of 2025 it has not been demonstrated beyond two species, but it represents the current state of the art from the CEA group for stochastic CD.

The BibTeX entry for this paper is:

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@article{Vasav2025,
  author  = {Vasav, Rohit and Jourdan, Thomas and Adjanor, Gilles and
             Badahmane, Achraf and Creuze, Jérôme and Athènes, Manuel},
  title   = {Cluster dynamics simulations using stochastic algorithms with
             logarithmic complexity in number of reactions},
  journal = {Journal of Computational Physics},
  year    = {2025},
  doi     = {10.1016/j.jcp.2025.114039},
}
```

6 Summary and Conclusions

Table 1 summarises the landscape of methods and their demonstrated multispecies reach.

Table 1: Summary of hybrid and stochastic cluster dynamics methods and their demonstrated maximum number of co-clustering species. "V+He" denotes a helium–vacancy two-species system; "V+He+H" denotes the full triple-species fusion irradiation problem.

Method	**Group**	**Max species**	**Demonstrated application**
Terrier 2017 Lie–Trotter hybrid [1]	CEA/CERMICS	1 (V only)	Thermal ageing, proof of concept
Jourdan 2014 det. FP coupling [6]	CEA/EDF	2 (V + He)	He implantation, annealing in α-Fe
SCD — Marian & Bulatov 2011 [7]	UCLA/LLNL	3 (V + He + H)	Triple-beam Fe irradiation to 1 dpa
Enhanced SCD — Hoang et al. 2015 [8]	UCLA/LLNL/UCB	3 (V + He + H)	Triple-beam Fe thin films, high dose
Log-complexity SSA — Vasav et al. 2025 [9]	CEA	2 (V + Cu)	FeCu precipitation under irradiation

The main conclusions are:

1. **The Terrier 2017 hybrid has never been applied to more than one cluster species.** Its proof of concept is single-species (vacancy-only) thermal ageing. The method was designed with multispecies extension in mind (the stochastic large-cluster step scales with particle count, not species count), but that extension remains unpublished.

2. **The deterministic FP–rate-equation coupling (Jourdan 2014) reaches two species** (V + He) and is the production basis for the CRESCENDO code used at CEA/EDF. The FP equation becomes a multi-dimensional mesh problem for three or more species and is not tractable beyond two.

3. **The only confirmed three-species CD applications are via pure SCD** (Marian & Bulatov 2011; Hoang et al. 2015), which handle the V + He + H triple-beam problem at doses up to and beyond 1 dpa. SCD avoids the ODE size problem by operating on a finite population in a finite volume, at the cost of statistical noise and the inability to resolve spatial correlations.

4. **Radiation-induced precipitation with four or more species** (e.g., the Mn–Ni–Si–Cu problem in RPV steels) has not been demonstrated by any CD-family method. It remains an open computational challenge, with OKMC being the current tool of choice despite its severe dose limitations.

5. **The most recent advance** is the logarithmic-complexity SSA of Vasav et al. [9], which offers orders-of-magnitude speedup over the standard Gillespie algorithm and is explicitly designed as a stepping stone to multispecies simulations, though it has currently been demonstrated only on a two-species (V + Cu) problem.

For the EUROFER97 CD framework described in this document — which involves V, I, and He as the three primary cluster species — the most practical path to a multispecies simulation appears to be either:

- Implementing the He reductions (Case 1 or Case 2) described in the He-content reduction section to collapse the He dimension, reducing the problem to effectively two species (SIA and VacHe), which is tractable by the deterministic approach with logarithmic size grouping; or
- Adopting the SCD framework of Marian & Bulatov [7], extended with τ-leaping [8], which can handle all three species simultaneously at the cost of stochastic noise and loss of the per-cluster-size resolution that the deterministic framework provides.

The Terrier hybrid in principle offers the best of both worlds for a future three-species implementation, but the work of extending the splitting formulation to multiple mobile monomers (both c_1^v and c_1^i and c_h as monomer concentrations governed by coupled nonlinear ODEs) remains to be done.

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